Notes on Algebraic Number Theory

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Contents

0. Introduction

This is a set of notes I am writing to help me study for Professor Tomer Schlank's course on Algebraic Number theory. Since these are notes (not a textbook), the main goal is just to speed through all the content as fast as possible so I can remember everything easily.

1. Commutative Algebra

In this section I will speed through a lot of the basic commutative algebra and background knowledge needed to do Algebraic Number theory.

1.1. Fractional Ideals

Definition 1.1 (Fractional Ideals)**.** Let R be an integral domain and K its field of fractions. A *fractional ideal* I is an R-submodule of K such that there exists some $r \in R$ with $rI \subseteq R$ We call I *invertable* if there exists a fractional ideal J with $IJ = R$.

In general it is hard to understand multiplication in interesting rings. However if we can instead understand multiplication in the *group of fractional ideals* we begin making progress.

Theorem 1.2.

(a) If I is an invertable fractional ideal with $IJ = R$ then $J = R :_K I$.

(b) We have the implications

$$I \text{ non-zero principal} \quad \Rightarrow \quad I \text{ invertable} \quad \Rightarrow \quad I \text{ fractional}$$

Proof.

(a) By the hypothesis that I is invertable we see that $(R :_K I)$ must work as an inverse. Therefore it suffices to prove that inverses are unique. Suppose J, J' invert I and write:

$$J = JR = J(J'I) = J'(JI) = J'$$

(b) A principal R-submodule of K, (x) has inverse (x^{-1}). If I is an invertable fractional ideal then any element $x \in (R :_K I)$ must have $xI \subseteq R$ as desired.

□

In general for an R-submodule of K we will denote $(R :_K I)$ as I^{-1}.

Theorem 1.3. *Any finitely generated R-submodule of K is a fractional ideal. Moreover if R is Noetherian then the opposite implication holds.*

Proof.

⇒: Suppose that $I = (a_1/b_1, \dots, a_n/b_n)$ for $a_i, b_i \in R$. Then if $x = b_1 \cdots b_n$ we see $xI \subseteq R$.

⇐: By assumption we may find x with $xI \subseteq R$. Then by the Noetherian condition $xI = (a_1, \dots, a_n)$ so $I = (a_1/x, \dots, a_n/x)$ as desired.

□

Theorem 1.4. *Let I an invertable fractional ideal of R. Then I is finitely generated, and moreover if R is local with unique maximal ideal $\mathfrak{m}$ then I is principal.*

Proof. Suppose that $II^{-1} = R$. Then since $1 \in II^{-1}$ we may find $a_i \in I$ and $b_i \in I^{-1}$ so that

$$\sum_{i=1}^{n} a_i b_i = 1$$

and $\langle a_1, \ldots, a_n \rangle I^{-1} = R$. However by Theorem 1.2 I is the unique inverse to I^{-1} so $I = \langle a_1, \ldots, a_n \rangle$ is finitely generated. In the case that R is local we take $\mod \mathfrak{m}$ we see at least one of $a_k b_k$ is not in $\mathfrak{m}$. However since R is local this means that $a_k b_k = u$ is a unit. Therefore $\langle a_k \rangle I^{-1} = R$ so by the same argument $I = \langle a_k \rangle$. □

1.2. Discrete Valuation Rings

Definition 1.5 (Valuation)**.** A *valuation* on a ring R is a mapping $v : K \to G \cup \{\infty\}$, where G is a totally ordered abelian group, satisfying:

(a) $v(a) = \infty \iff a = 0$

(b) $v(ab) = v(a) + v(b)$

(c) $v(a + b) \geq \min\{v(a), v(b)\}$

(d) The image of v is nontrivial

We call v a *discrete valuation* if $G \cong \mathbb{Z}$.

It is worth noting some people do not enforce the last condition. We mainly do it to disallow the trivial valuation $v(R \setminus \{0\}) = 0$. To give some intuition for a valuation, here are some familiar examples:

- The p-adic valuation on $\mathbb{Q}$. For a prime p, $v_p(x)$ is the exponent of the highest power of p dividing x.
- The a-adic valuation on $k(x)$. For $a \in k$, $v_a(f)$ is the order of the zero (if positive) or pole (if negative) of f at $x = a$.
- The degree valuation on $k(x)$. Defined as $v_\infty(P/Q) = \deg(Q) - \deg(P)$.
- The Laurent series valuation. On $k((t))$, $v(f)$ is the smallest n such that the coefficient a_n of t^n is non-zero.
- The monomial valuation on $k(x, y)$. For a fixed weight $w = (w_1, w_2)$, $v(\sum a_{ij} x^i y^j) = \min\{iw_1 + jw_2 : a_{ij} \neq 0\}$.

In general, thinking of a valuation as a way to count the order of a zero or a pole of a function on a point gives very good intuition. The reason we care about valuations is their connection to the concept of a valuation ring.

Definition 1.6 (Valuation Rings)**.** We call a ring R with fraction field K a *valuation ring* if one of the two equivalent conditions is satisfied:

(a) For every $x \in K$ either x or x^{-1} is in R.

(b) There exists a valuation on K such that $R = \{x \in K : v(x) \geq 0\}$

Proof.

$(a) \Rightarrow (b)$**:** We define the valuation group $G = R/R^\times$ to be the multiplicative group of R quotient the units. The valuation is $v(r) = \bar{r}$ and $v(0) = \infty$. We need to show three things:

G **is totally ordered:** We claim that the ideals of R are totally ordered by inclusion. This induces a total order on G by $\bar{a} > \bar{b}$ if $(a) \subset (b)$. Suppose FTSOC that I and J are ideals such that we can find some $x \in I \setminus J$ and $y \in J \setminus I$. By assumption either x/y or y/x is in R, so suppose WLOG that $x/y \in R$. Then $x = y \cdot (x/y)$ which implies tht $x \in (y) \subseteq I$, a contradiction.

$v(ab) = v(a) + v(b)$**:** This follows directly from the definitions.

$v(a+b) \geq \min(v(a), v(b))$**:** This translates to the statement that

$$(a+b) \subseteq \max((a), (b)) = (a, b)$$

where the equality uses the fact that either $(a) \subseteq (b)$ or $(b) \subseteq (a)$.

$(b) \Rightarrow (a)$**:** For $x \in K$ we either have $v(x) \geq 0$ or $v(x) \leq 0$. If $v(x) \geq 0$, then $x \in R$. If $v(x) \leq 0$, then $v(x^{-1}) = -v(x) \geq 0$, so $x^{-1} \in R$.

□

Definition 1.7 (Discrete Valuation Ring)**.** We call a valuation ring *discrete* if its valuation is discrete.

The concept of a discrete valuation ring (often abreviated as DVR) is extremely important in algebraic number theory. Therefore try very hard to internalize the concept via the following examples:

- The ring of p-adic integers, denoted $\mathbb{Z}_p$. The uniformizer is the prime number p, and the unique maximal ideal is $p\mathbb{Z}_p$.
- Formal power series rings, denoted $k[[t]]$ where k is a field. The uniformizer is the variable t. Elements are of the form $\sum_{n=0}^{\infty} a_n t^n$.
- The localization of $\mathbb{Z}$ at a prime p, denoted $\mathbb{Z}_{(p)}$. This consists of rational numbers a/b where $p \nmid b$. The uniformizer is p.
- The local ring of a curve at a smooth point. If C is a non-singular algebraic curve and $P \in C$ is a point, the local ring $\mathcal{O}_{C,P}$ is a DVR. The uniformizer is any local parameter at P.

It is worth noting that a ring having a valuation does not make it a valuation ring. However if R has a valuation, then $\mathfrak{p} = \{r \in R : v(r) > 0\}$ is a prime ideal of the subring $R' = \{r \in R : v(r) \geq 0\}$. It turns out that the localization $R'_{\mathfrak{p}}$ is a valuation ring with $v(a/b) = v(a) - v(b)$.

Problem 1.8. *Prove that a discrete valuation ring is a Noetherian local domain*

Now we have some characteriziations of a discrete valuation ring.

Theorem 1.9. *For a Noetherian local ring R the following statements are equivalent:*

(a) R is a discrete valuation ring.

(b) R is principal ideal domain, but not field.

(c) R is a 1-dimensional unique factorization domain.

(d) R is a 1-dimensional normal domain.

(e) R is a 1-dimensional regular ring.

Proof. Through the proofs we denote $\mathfrak{m}$ as the unique maximal ideal of R.

$(a) \Rightarrow (b)$**:** By the Noetherian condition every ideal can be written as $I = (a_1, \ldots, a_n)$. Then $I = (a_i)$ where a_i is the generator with minimal valuation. Since the valuation is nontrivial there must be non-unit element.

$(b) \Rightarrow (c)$**:** Since a PID is a UFD we only have to show that R is 1-dimensional. Suppose FTSOC that there is a prime $\mathfrak{p} \subsetneq \mathfrak{m}$. We can write $p = (\rho)$, $\mathfrak{m} = (\pi)$ and find some element x with $x\pi = \rho$. However since ρ generates a prime ideal it is irreducible, a contradiction.

$(c) \Rightarrow (d)$**:** It is a general result that every UFD is normal.

$(d) \Rightarrow (e)$**:** We will show that $\mathfrak{m}$ is principle by showing it is invertable, so consider $\mathfrak{m}\mathfrak{m}^{-1}$ (where $\mathfrak{m}^{-1} := (R :_K \mathfrak{m})$). By definition this is an ideal in R containing $\mathfrak{m}$, therefore it is either $\mathfrak{m}$ or R.

$\mathfrak{m}\mathfrak{m}^{-1} = \mathfrak{m}$**:** In this case every element $x \in \mathfrak{m}^{-1}$ satisfies $x\mathfrak{m} \subseteq \mathfrak{m}$ which implies by Nakayama lemma that x is integral over R and therefore in R by assumption. Pick some $x \in \mathfrak{m}$ and by Krull intersection theorem let n be minimal so that $\mathfrak{m}^n \subseteq (x)$. Therefore $x^{-1}\mathfrak{m}^n \in R$, however $x^{-1}\mathfrak{m}^{n-1} \not\subseteq R$. Then any element $y \in x^{-1}\mathfrak{m}^{n-1} \setminus R$ satisfies $y\mathfrak{m} \subseteq R$ and $y \notin R$, a contradiction.

$\mathfrak{m}\mathfrak{m}^{-1} = R$**:** In this case $\mathfrak{m}$ is an invertable ideal. Now by Theorem 1.4 every invertable ideal is principal as desired.

$(e) \Rightarrow (a)$**:** By Nakayama $\dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = 1$ implies that $\mathfrak{m}$ is principal with some generator π. We claim that every element $x \in R$ is of the form $u\pi^n$ for $u \in R^\times$, at which point it follows that R is a DVR. By Krull intersection theorem let n be maximal so that $x \in (\pi^n)$ so $x = y\pi^n$. By maximality of n we see $y \notin (\pi)$, and since π is the unique maximal ideal this means y is a unit.

□

These might seem hard to remember, but the last 3 conditions essentially say the same thing. That a DVR looks like the functions near a smooth point on a curve. The slogan is that for a *curve* the following are the same:

- There is *one* codimension-1 direction through a smooth point.
- Near a smooth point there is a single local coordinate (a *uniformizer*).
- Every function has a well-defined *order of vanishing* which is just an integer.

This is exactly the discrete valuation story. Geometrically, the last three conditions in Theorem 1.9 are three different ways of saying

"near P the curve looks like one smooth branch, i.e. like a line."

In dimension 1 these points of view collapse; in higher dimension they genuinely diverge.

Here is a useful mental dictionary for a local ring $R = \mathcal{O}_{C,P}$ of an integral curve C at a point P:

- *Regular (smoothness / tangent space).* At a smooth point you can choose one local parameter t cutting out P. Algebraically this means the maximal ideal is generated by one element:

  $$\mathfrak{m}_P = (t),$$

  and infinitesimally there is only one direction along the curve:

  $$\dim_{\kappa(P)}(\mathfrak{m}_P/\mathfrak{m}_P^2) = 1.$$

  A singular point forces at least two independent first-order directions (or "too many tangent vectors"), which is why $\mathfrak{m}_P$ stops being principal.

- *Normal (no hidden functions / normalization).* The normalization $\nu : \widetilde{C} \to C$ is the operation that "pulls apart" branches and "fills in" missing integral functions. Saying R is normal means that locally nothing changes under normalization: there are no missing integral elements. On a curve, the only way to fail normality is precisely to have a singularity like a cusp or a node: there is some rational function that is integral over R which should be considered regular on the curve, but is not actually in R.

- *UFD (divisors become a single integer).* On a smooth curve, any non-zero function f has a well-defined integer $\mathrm{ord}_P(f)$ measuring how many times it vanishes at P (negative if it has a pole). The key point is that for a smooth point this is *one integer*, not a vector of integers. This corresponds to the fact that every ideal is just a power of (t):

  $$I \subseteq R \iff I = (t^n) \text{ for a unique } n \geq 0.$$

  So locally there is a single "prime divisor" $[P]$, and everything factors as a multiple of it. If the point has multiple branches, you would instead get a *tuple* of orders (one per branch), and there is no longer a single generator controlling all vanishing.

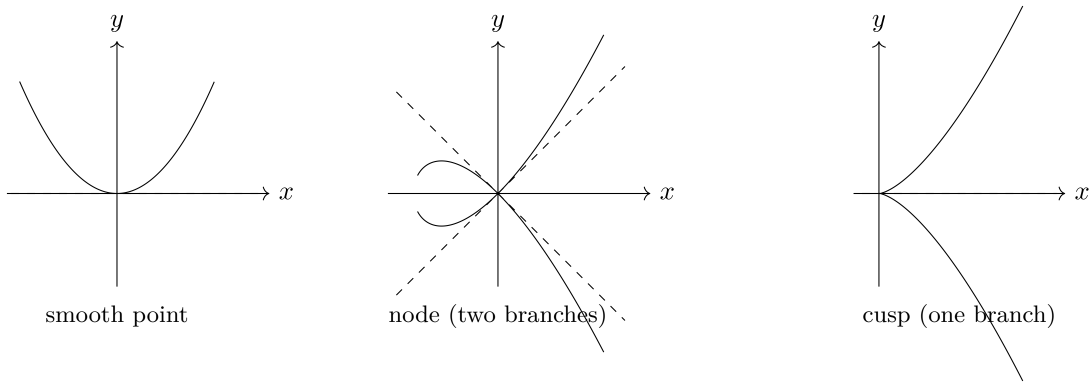

Figure 1: Singularities force "too many" first-order directions. At a smooth point there is one local parameter t; at a node there are two branches; at a cusp there is one branch but the tangent is "thick". In both singular pictures the maximal ideal cannot be generated by one element, and normalization changes the local ring.

It is helpful to keep two toy local models in mind (purely as intuition):

- *Cusp: one branch but not normal.* The parametrization $t \mapsto (t^2, t^3)$ suggests that (after completing) the local ring looks like
$$k[[t^2, t^3]] \subsetneq k[[t]].$$
The element t is "the missing uniformizer": adjoining it is exactly what normalization does. This is why normality fails, and consequently regularity/UFD fail as well.

- *Node: two branches.* Near a node there are two ways to approach P along the curve, so vanishing wants to be measured by
$$(\mathrm{ord}_{P,1}(f),\ \mathrm{ord}_{P,2}(f)) \in \mathbb{Z}^2$$
rather than a single integer. Normalization separates the branches, and locally you should think of it as producing (roughly) two DVRs, one for each branch. The original local ring is their intersection inside the function field, which explains why it is not itself a DVR.

So, on a curve, *regular* says "one parameter" (one tangent direction), *normal* says "no hidden integral functions" (normalization does nothing), and *UFD* says "orders of vanishing are controlled by a single integer." These are three shadows of the same geometric phenomenon.

1.3. Dedekind Domains

A DVR is a very local notion, so you might wonder if there is corresponding global notion. The answer is yes, and it is a Dedekind domain. Just like with a DVR, there are many equivalent conditions for a ring to be a Dedekind domain. Now buckle up, because we're about to start making the rounds.

Definition 1.10 (Dedekind Domain)**.** We call an integral domain R that is not a field a Dedekind domain if it satifies any of the equivalent conditions:

(a) R is a Noetherian, and the localization at every maximal ideal is a discrete valuation ring.

(b) R is a normal (or regular), 1-dimensional Noetherian domain.

(c) Every quotient of R is a principal ideal domain.

(d) Every nonzero proper ideal factors into primes.

(e) Every nonzero fractional ideal of R is invertable.

(f) For any ideals I, J in R, I is contained in J if and only if J divides I. That is, there is an ideal H with $I = JH$.

Proof.

$(a) \Leftrightarrow (b)$**:** Firstly, (a) implies that R is 1-dimensional since otherwise localizing at a maximal ideal above a nontrivial prime does not give a DVR. Recall from Theorem 1.9 that a DVR is a Noetherian, 1-dimensional, normal/regular *local* ring. Therefore we only need to show that normality and regularity are local properties, at least when every prime is maximal.

- Note that there is the standard inclusion map $\psi : R \to \bar{R}$ where $\bar{R}$ is the integral closure of R in its field of fractions. By commutative algebra ψ is an isomorphism iff $\psi_{\mathfrak{m}}$ is an isomorphism for all $\mathfrak{m} \in \operatorname{Spm} R$. However since integral closure commutes with localization this is exactly the condition that $R_{\mathfrak{m}}$ is integrally closed for all $\mathfrak{m}$.
- Regularity means that $\dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = 1$ for every $\mathfrak{m} \in \operatorname{Spm} R$. However $R/\mathfrak{m}$ is local, and in particular $(R/\mathfrak{m})_{\mathfrak{m}} = R/\mathfrak{m}$ so this is a local condition.

$(f) \Rightarrow (e)$**:**

$(e) \Rightarrow (a)$**:** Let $\mathfrak{m}$ be a maximal ideal of R and note that

$$I_{\mathfrak{m}} I_{\mathfrak{m}}^{-1} = (II^{-1})_{\mathfrak{m}} = R_{\mathfrak{m}}$$

so the localization retains the property of every ideal being invertable. Therefore $R_{\mathfrak{m}}$ is a PID by Theorem 1.4, so by Theorem 1.9 (a) we are done.

$(a) \Rightarrow (d)$**:** Suppose that I is an ideal of R. Since R is Noetherian so by the Noether-Lasker theorem we may take a primary decomposition $I = Q_1 \cap \cdots \cap Q_n$. We want to show that $Q_i = \mathfrak{p}_i^k$ for the prime ideal $\mathfrak{p}_i = \sqrt{Q_i}$. We denote $S = R \setminus \mathfrak{p}$ and note:

$$(Q_i)_{\mathfrak{p}} \cap R = \{x \in R | (Q : x) \cap S \neq \emptyset\}$$

However by the primary condition if $sx \in Q_i$ for $s \notin \mathfrak{p}$ then $x \in Q_i$. Therefore $(Q_i)_{\mathfrak{p}} \cap R = Q_i$. Since $R_{\mathfrak{p}}$ is a DVR we see that $(Q_i)_{\mathfrak{p}} = \mathfrak{p}^k$ implying that $Q_i = \mathfrak{p}^k$. Since every prime is maximal by (b) we can use CRT to write this as $I = \mathfrak{p}_1^{k_1} \cdots \mathfrak{p}_n^{k_n}$

$(d) \Rightarrow (f)$**:** We claim that every prime ideal is maximal. Suppose FTSOC that $0 \subsetneq \mathfrak{p} \subsetneq \mathfrak{q} \subsetneq R$. We consider the ideal $I = \mathfrak{p} + \mathfrak{q}^2 = \mathfrak{r}_1 \cdots \mathfrak{r}_n$ for $\mathfrak{r} \in \operatorname{Spec} R$ by the prime factorization property. By primality $\mathfrak{q}$ contains at least one of these ideals, say $\mathfrak{r}_1$.

□

In general the standard way to check whether a ring is a Dedekind domain is via (b). That is that a Dedekind domain is normal 1-dimensional Notherian domain. One may think of (a)-(c) as expressing the notion that a Dedekind domain looks locally like a DVR. On the other hand (d)-(f) express the fact that the fractional ideals behave like a group. The connection is in algebraic geometry:

$$\text{fractional ideals} \quad \cong \quad \text{line bundles} \quad \cong \quad \text{divisors}$$

so being locally a DVR (smooth curve) means that divisors are just sums of points. As before, here are some examples of a Dedekind domain:

- The ring of integers $\mathbb{Z}$ is a principal ideal domain and the archetypal example of a Dedekind domain.
- The polynomial ring $k[x]$ over any field k is a Euclidean domain and therefore a Dedekind domain.
- The ring of integers $\mathcal{O}_K$ of a number field K (a finite extension of $\mathbb{Q}$) is a Dedekind domain.
- The Gaussian integers $\mathbb{Z}[i]$ form a principal ideal domain, which implies they are a Dedekind domain.
- Any discrete valuation ring (DVR), such as the ring of p-adic integers $\mathbb{Z}_p$, is a local Dedekind domain.

- The coordinate ring $k[C]$ of a smooth (non-singular) affine algebraic curve C over an algebraically closed field is a Dedekind domain.
- The ring $\mathbb{Z}[\sqrt{-5}]$ is a Dedekind domain that is not a principal ideal domain, illustrating that element factorization need not be unique even when ideal factorization is.

Now we can define the main exact sequence of study in algebraic number theory.

Construction 1.11. Let $\mathscr{O}_K$ be a Dedekind domain with fraction field K. We define the following objects:

$\mathcal{I}$**:** The *ideal group* $\mathcal{I}$ is the group of fractional ideals under multiplication.

Cl_K**:** The *class group* $\mathrm{Cl}_K := \mathcal{I}_K/\mathcal{P}_K$ is the quotient of $\mathcal{I}_K$ by the *principal divisors* $\mathcal{P}_K$ which are the group of principal fractional ideals.

Then there is the following exact sequence

$$1 \to \mathscr{O}_K^\times \to K^\times \to \mathcal{I}_K \to \mathrm{Cl}_K \to 1 \tag{1}$$

Proof. We will show exactness at each point.

- The map $\mathscr{O}_K^\times \to K^\times$ is by inclusion and the map $K^\times \to \mathcal{I}_K$ is by $x \mapsto (x)$. The elements which generate $\mathscr{O}_K$ (the zero element in $\mathcal{I}_K$) are exactly the units.
- The map $\mathcal{I} \to \mathrm{Cl}_K$ is the quotient map, and the kernel are the principal ideals. These are exactly the ideals of the form (x) for $x \in K^\times$.

□

Remark 1.12. Note that

$$\mathcal{I}_K \cong \bigoplus_{\mathfrak{p} \in \mathrm{Spec}\, \mathscr{O}_K} \mathbb{Z}$$

Proof. Any fractional ideal can be characterized by the power of each prime appearing in the decomposition. □

Remark 1.12 gives us a very good understanding of the ideal group. We will eventually better understand $\mathscr{O}_K^\times$ and Cl_K so that we may understand multiplication in $K^\times$ and $\mathscr{O}_K$ via the exact sequence in Eq. (1).

2. How Primes Behave

In this section you will learn how to determine what primes do in extensions of Dedekind domains (such as number fields).

2.1. The Trace Pairing

In this section I will define the discriminant of a field extension, and how it comes from the trace pairing.

Definition 2.1 (Bilinear Form)**.** A k-linear *bilinear form* on a vector space V is a bilinear map $V \times V \to k$. If V is finite dimensional, we define the *discriminant* of a bilinear form relative to a basis e_i as $\det\langle e_i, e_j\rangle$. We call a bilinear form *non-degenerate* or *perfect* if it satisfies the following equivalent conditions:

(a) it has nonzero discriminant with respect to one (hence every) basis;

(b) for every $x \in V$ there is a $y \in V$ with $\langle x, y\rangle \neq 0$;

(c) for every $y \in V$ there is a $x \in V$ with $\langle x, y\rangle \neq 0$.

Proof. First we prove the claim that (a) does not care about basis. For a basis $e_1, \dots, e_n$ we let A denote the matrix $\langle e_i, e_j \rangle$. Then

$$\langle x, y \rangle = x^T A y$$

which can be verified by writing x and y in the e_i basis. Another basis $f_1, \dots, f_n$ induces an invertable change of basis matrix M with $Me_i = f_i$. Then

$$\langle f_i, f_j \rangle = \langle Me_i, Me_j \rangle = e_i^T M^T A M e_j$$

and the discriminant with respect to this basis is $\det M^T A M = \det A \cdot (\det M)^2$ which is nonzero iff $\det A$ is nonzero. Now it suffices by symmetry to prove $(a) \Leftrightarrow (c)$.

$(a) \Rightarrow (b)$: For some $x \in V$ we consider $y = A^{-1}x$ so that

$$\langle x, y \rangle = x^T A y = x^T x$$

which is always nonzero.

$(b) \Rightarrow (a)$: Suppose in the contrapositive that $\det A = 0$. Then there is some $y \in V$ with $Ay = 0$. It is now clear there is no $x \in V$ with $x^T A y \neq 0$.

□

Note that the discriminant of a bilinear form is defined up to the square of a number, due to the change of basis formula.

Remark 2.2. A good way to think about the discriminant is the following. A bilinear form $\langle \rangle$ gives some way to measure distance, angles, and volume. The discriminant of a basis $e_1, \dots, e_n$ is exactly the square of the covolume of the lattice it generates.

Proof. The discriminant with respect to an orthonormal basis $\langle e_i, e_j \rangle = \delta_{ij}$ is 1. Then changing the basis scales the discriminant by the square of the determinant of the change of basis matrix. □

Definition 2.3 (R-lattice)**.** Let R be an integral domain with fraction field K, V a finite dimensional K-vector space, and M and R-submodule of V. We call a set of elements $\{\beta_1, \dots, \beta_n\}$ a *basis* for M if they generate M over R and are simultaneously a basis for V over K. M is called an *R-lattice* if it has a basis. We define the *discriminant* of M/R to be the discriminant of the trace pairing with respect to a basis of M.

Note that $\operatorname{disc} V/K$ is an element of $K/K^{\times 2}$ (defined up to the square of an element) by our analysis in Definition 2.1. The field discriminant does not carry much information. On the other hand two bases of M over R must be related by an R-matrix, and $\operatorname{disc} M/R$ is an therefore an element of $R/R^{\times 2}$. This allows it to carry much more information since R might not have that many units. The most important example of this is when $R = \mathbb{Z}$ and M is the ring of integers in finite separable extension of $\mathbb{Q}$.

Lemma 2.4. *Recall that the trace pairing defines an isomorphism $\tau_{V/R} : V \to V^\vee$ which identifies $M^\vee$ with $M^\sharp$ inside V. Then the discriminant is (up to units) the determinant of any lattice map sending $M^\sharp$ to M.*

Proof. Let $e_1, \dots, e_n$ be a basis for M and $e_1^*, \dots, e_n^*$ be the dual basis generating $M^\sharp$. Since

$$e_i = \sum_j \langle e_i, e_j \rangle e_j^*$$

the matrix $\langle e_i, e_j \rangle$ is the change of basis matrix from $e_1, \dots, e_n$ to $e_1^*, \dots, e_n^*$. Therefore as a matrix in the dual basis it sends $M^\sharp$ to M. On the other hand the determinant is $\operatorname{disc} M/R$ by defintion. □

Construction 2.5. Let R be an integral domain with fraction field K, V a finite dimensional K-vector space, and $\langle \rangle$ a perfect pairing $V \times V \to K$. For any R-submodule M of V, we can define the *dual* of M as:

$$M^\sharp := \{x \in L : \langle x, y \rangle \in R \text{ for all } y \in M\}$$

Suppose moreover that M is an R lattice. Then $M^\sharp$ is an R-lattice generated by the *dual basis*.

Proof. Since the pairing is perfect it induces an isomorphism (via dimension counting)

$$V \xrightarrow{\sim} V^\vee \qquad x \mapsto \langle x, - \rangle$$

and $M^\sharp$ is the preimage of $M^\vee = \mathrm{Hom}_R(M, R)$ (sitting inside $V^\vee = \mathrm{Hom}_K(V, K)$) under this isomorphism. Suppose that $e_1, \ldots, e_n$ is an R-basis of M that is also a K-basis of V, we can use this isomorphism to induce a unique *dual basis* $e_1^*, \ldots, e_n^*$ which has the property that:

$$\langle e_i, e_j^* \rangle = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

We claim that this dual basis $e_1^*, \ldots, e_n^*$ generates $M^\sharp$. Suppose that $x \in M^\sharp$ and write $x = x_1 e_1^* + \cdots + x_n e_n^*$ for $x_i \in K$. By the assumption that $x \in M^\sharp$ we see $\langle x, e_i \rangle = x_1$ is in R as desired. □

Geometrically suppose that $R = \mathbb{Z}$ in Construction 2.5. Then the conditions we put on M exactly translate to it being a *lattice* in $\mathbb{Q}^n$ with full rank. The theorem tells us that the set of points whose pairing with the lattice gives only integers is another lattice. The *dual lattice* if you will. This geometric picture is very important to keep in mind. In fact, we call an M satisfying the conditions in Construction 2.5 an R-lattice.

Definition 2.6. Suppose that L/K is a finite extension, then L has the structure of a K vector space. Therefore any element $x \in L$ induces a K-linear transformation on L by multiplication by x. We can then define:

- the *norm* $\mathrm{N}_{L/K}(x)$ to be the determinant of the transformation;
- the *trace* $\mathrm{Tr}_{L/K}(x)$ to be the trace of the transformation.

Definition 2.7 (Trace Pairing)**.** The *trace pairing* of a finite extension L/K is the bilinear form:

$$\langle x, y \rangle = \mathrm{Tr}(xy)$$

The *discriminant* of the field extension is the discriminant of this pairing.

The trace pairing is the most natural bilinear form to use, so from now on we will let $\langle x, y \rangle$ denote $\mathrm{Tr}(xy)$.

Proposition 2.8. *Suppose that L/K is a degree n extension and that $\beta_1, \ldots, \beta_n$ is a basis for L over K. Then*

$$\langle \beta_i, \beta_j \rangle = VV^T \qquad V := \begin{bmatrix} 1 & \cdots & 1 \\ \sigma_1(\beta_1) & \cdots & \sigma_n(\beta_1) \\ \vdots & & \vdots \\ \sigma_1(\beta_n) & \cdots & \sigma_n(\beta_n) \end{bmatrix}$$

where V is the embedding matrix and the σ_i are the n embeddings of L into the algebraic closure of K. In particular the discriminant of L/K is $(\det V)^2$*.*

Proof. We note that:

$$(VV^T)_{ij} = \sum_{k=1}^n V_{ik} V_{jk} = \sum_{k=1}^n \sigma_k(\beta_i)\sigma_k(\beta_j) = \mathrm{Tr}(\beta_i \beta_j)$$

□

Corollary 2.9. *The trace pairing of a finite extension L/K is non-degenerate if and only if the extension is separable.*

Proof. By Proposition 2.8 it suffices to consider when $\det V$ is nonzero.

- If L/K is *separable*, the embeddings σ_i are distinct, so by Dedekind's theorem on the linear independance of characters $\det V \neq 0$.

- If L/K is *inseparable*, then $\sigma_i = \sigma_j$ for some $i \neq j$. This makes two columns of V the same so $\det V = 0$.

$\square$

Example 2.10 (The Discriminant of a Polynomial and its Field)**.** Let $f(x) \in R[x]$ be a monic irreducible polynomial of degree n with roots $\alpha_1, \ldots, \alpha_n$ in a splitting field L over $K = \mathrm{Frac}(R)$. There are three ways we may define Δ_f (the discriminant of f):

Root Differences: Defined essentially as the product of squared differences of the roots. This vanishes if and only if roots collide (i.e., f is inseparable).

$$\Delta_f = \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)^2$$

The Resultant: Calculable without knowing the roots, using the resultant of f and its derivative f':

$$\Delta_f = (-1)^{\frac{n(n-1)}{2}} \mathrm{Res}(f, f')$$

We still need to define the *resultant* that we have used in the formula.

Let $f(x) = a_n x^n + \cdots + a_0$ and $g(x) = b_m x^m + \cdots + b_0$ be polynomials in $F[x]$ of degrees n and m respectively. The *Sylvester matrix* of f and g is the $(n+m) \times (n+m)$ matrix:

$$S_{f,g} = \begin{pmatrix} a_n & a_{n-1} & \ldots & a_0 & 0 & \ldots & 0 \\ 0 & a_n & \ldots & a_1 & a_0 & \ldots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \ldots & 0 & a_n & \ldots & \ldots & a_0 \\ b_m & b_{m-1} & \ldots & b_0 & 0 & \ldots & 0 \\ 0 & b_m & \ldots & b_1 & b_0 & \ldots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \ldots & 0 & b_m & \ldots & \ldots & b_0 \end{pmatrix}.$$

Now $\mathrm{Res}(f, g)$, is defined as the determinant of $S_{f,g}$, or alternatively as

$$\mathrm{Res}(f, g) = a_0^e b_0^d \prod_{\substack{1 \leq i \leq n \\ 1 \leq j \leq m}} (\lambda_i - \mu_j)$$

where λ_i and μ_j are the roots of f and g respectively.

Derivative Values: Expresses Δ_f via the values of the derivative at the roots.

$$\Delta_f = (-1)^{\frac{n(n-1)}{2}} \prod_{i=1}^{n} f'(\alpha_i) = (-1)^{\frac{n(n-1)}{2}} n^n \prod_{i=1}^{n-1} f(\beta_i)$$

where the β_i are the roots of f'.

If $L = K(\alpha)$ is the field generated by a root α of f, the discriminant of the power basis $\{1, \alpha, \ldots, \alpha^{n-1}\}$ (and hence $\operatorname{disc} R[\alpha]/R$) is exactly Δ_f.

Proof. First suppose that $K = F(\alpha)$ and f is the minimal polynomial of α. We may use the power basis $\{1, \alpha, \ldots, \alpha^{n-1}\}$ to compute the discriminant. By Proposition 2.8 this is the square of the determinant of the embedding matrix $V_{ij} = \sigma_i(\alpha^{j-1})$. This matrix is precisely the Vandermonde matrix of the roots $\alpha_1, \ldots, \alpha_n$ of f. Thus:

$$\Delta(1, \ldots, \alpha^{n-1}) = (\det V)^2 = \prod_{i<j} (\alpha_i - \alpha_j)^2 = \Delta_f.$$

which connects the field discriminant to at least one of our definitions.

The reader should show all the equivalences as an exercise (the last one may be too hard) or consult a standard text. Intuitively they all are different ways you might try to test whether f and f' share roots.

- The product $(\alpha_i - \alpha_j)^2$ computes whether f has a double root, which is true if and only if f and f' share a root.
- If f and f' share roots then by Bezout's theorem there are polynomials a and b (found by Euclid's algorithm) with:
$$af + bf' = 0 \qquad \deg a < \deg f' \qquad \deg b < \deg f$$
This forces the determinant of the Sylvester matrix to be 0 since it corresponds exactly to a linear dependence among the rows.
- If f and f' share roots then evaluating one at the roots of the other will give 0.

$\square$

Remark 2.11. Computing huge resultants is generally hard, however there is a sleak algorithm to do it quickly in a Euclidean domain.

(a) Use the Hadamard bound to see that the resultant's value is bounded by the product of the Euclidean norms of the rows.

(b) Pick small primes p_i which multiply to a number bigger than the bound in (a).

(c) Note that to compute the resultant mod p, we just reduce our matrix mod p and work in R/p.

(d) Note the formula (coming from the Sylvester matrix)
$$\mathrm{Res}(f, g) = (-1)^{mn}\,\mathrm{lc}(g)^{m-\deg h} \cdot \mathrm{Res}(h, g)$$
where lc is the leading coefficient of g and h is the remainder of f after division by g. This lets us compute using the Euclidean algorithm, which is not very expensive in finite fields.

(e) Use CRT to find the actual resultant.

Now we will use the trace pairing to understand Dedekind domains.

Lemma 2.12. *Let R be a normal domain with fraction field K, and let S be the integral closure of R is a separable extension L of K. There exist free R-submodules M and $M^\sharp$ with:"*
$$M \subseteq S \subseteq M^\sharp$$

Proof. Since L/K is separable, the trace pairing $(x, y) \mapsto \mathrm{Tr}_{L/K}(xy)$ is non-degenerate. We may find a basis for L over K contained in S and let M be the R-module generated by this basis. Then Construction 2.5 tells us that
$$M^\sharp := \{x \in L : \langle x, y\rangle \in R \text{ for all } y \in M\}$$
is finitely generated by the dual basis. Note that
$$\mathrm{Tr}_{L/K}(SM) \subseteq \mathrm{Tr}_{L/K}(S) \subseteq R$$
since $M \subseteq S$ and the trace of something integral over R is in R (by normality). Therefore $S \subseteq M^\sharp$ by definition and $M \subseteq S \subseteq M^\sharp$ as desired. $\square$

2.2. The Dedekind-Kummer Theorem

You might ask why we just spent so much time talking about field theory, when understanding Dedekind domain is our main goal. That is due to the following theorem.

For the remainder let R be a Dedekind domain with fraction field K, let L/K be a finite separable extension of degree n, and let S be the integral closure of R in L.

Theorem 2.13. *S is a Dedekind domain.*

Proof. To prove that S is a Dedekind domain, we must verify three conditions:

- S is integrally closed in L. By definition, S is the integral closure of R in L. Since the integral closure of a ring in a field is always integrally closed, S is integrally closed in $L = \mathrm{Frac}(S)$.
- Every non-zero prime ideal of S is maximal. Let $\mathfrak{q}$ be a non-zero prime ideal of S, and let $\mathfrak{p} = \mathfrak{q} \cap R$. Since S is integral over R, $\mathfrak{p}$ is a non-zero prime ideal of R. Because R is a Dedekind domain, $\mathfrak{p}$ is maximal. By the properties of integral extensions (specifically, the *Going-up Theorem*), the maximality of $\mathfrak{p}$ implies the maximality of $\mathfrak{q}$. Thus, S has Krull dimension 1.
- S is Noetherian. Note that Lemma 2.12 implies that S is a submodule of a finitely generated module. Since R is Noetherian Hilbert's basis theorem implies that S is Noetherian.

□

Remark 2.14. In the case that R is a PID, we note $S \cong R^n$ as an R-module. This is by the structure theorem for finitely generated modules over a PID, and the observation that S is torsion-free. In this case the R-basis for S is also a K-basis for L. For linear independence, note that a linear dependence with coefficients in K leads to one with coefficients in R after clearing denominators. Now we need that $S \otimes K = L$, ie that S spans L. Note any $x \in L$ satisfies a monic polynomial with coefficients in K. By clearing denominators we can find $r \in R$ with rx satisfying a monic polynomial with coefficients in R. Therefore $rx \in S$ and S is an R-lattice.

Definition 2.15. For any prime $\mathfrak{p} \in \operatorname{Spec} R$ we may note

(a) $\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_k^{e_k}$;

(b) The $\mathfrak{q}_i$ are exactly the primes with $\mathfrak{q}_i \cap R = \mathfrak{p}$.

We call the e_i the *ramification degree*, we call $\dim_{R/\mathfrak{p}} S/\mathfrak{q}_i$ the *inertia degree* of $\mathfrak{q}_i$ and denote it as f_i. We call the number of primes g. There are the following special cases:

- if $f_i = e_i = 1$ then $\mathfrak{p}$ is *totally split*;
- if $f_i = g = 1$ then $\mathfrak{p}$ is *totally ramified*;
- if $e_1 = g = 1$ then $\mathfrak{p}$ is *inert*;
- if $e_i = 1$ then $\mathfrak{p}$ is *unramified*, otherwise it is *ramified*.

Proof.

(a) S is a Dedekind domain.

(b) Note that $\mathfrak{q}_i$ contains $\mathfrak{p}S$ so by the prime avoidance theorem it must contain one of the factors in the decomposition. Then since every prime is maximal it is one of the factors.

□

Theorem 2.16. *For any nonzero prime ideal $\mathfrak{p} \subset R$, let the factorization of $\mathfrak{p}S$ in S be given by $\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_g^{e_g}$, and let $f_i = [S/\mathfrak{q}_i : R/\mathfrak{p}]$ be the residue degree. Then:*

$$\sum_{i=1}^{g} e_i f_i = n$$

Proof. We analyze the dimension of the $R/\mathfrak{p}$-vector space $S/\mathfrak{p}S$ in two different ways.

S is Locally Free: First we claim that S is a projective R-module. It suffices to show that S is locally free. Let $\mathfrak{p}$ be any prime ideal of R. The localization $R_\mathfrak{p}$ is a DVR, and hence a PID. Then by Remark 2.14 $S_\mathfrak{p}$ is an $R_\mathfrak{p}$-lattice and $L \cong S \otimes_R K \cong S_\mathfrak{p} \otimes_{R_\mathfrak{p}} K$, the rank of this free module is:

$$\dim_K(S_\mathfrak{p} \otimes_{R_\mathfrak{p}} K) = [L : K] = n$$

Thus, S is locally free of rank n. This implies S is projective.

Dimension via Rank: We compute the dimension of $S/\mathfrak{p}S$ over the field $k = R/\mathfrak{p}$. Since S is locally free of rank n, we can localize at $\mathfrak{p}$ without changing the quotient:

$$S/\mathfrak{p}S \cong S_\mathfrak{p}/\mathfrak{p}S_\mathfrak{p} \cong S_\mathfrak{p}/\mathfrak{p}R_\mathfrak{p}S_\mathfrak{p}$$

Because $S_\mathfrak{p} \cong R_\mathfrak{p}^{\oplus n}$, we have:

$$S_\mathfrak{p}/\mathfrak{p}S_\mathfrak{p} \cong (R_\mathfrak{p}/\mathfrak{p}R_\mathfrak{p})^{\oplus n} \cong k^{\oplus n}$$

Therefore,

$$\dim_k(S/\mathfrak{p}S) = n$$

Dimension via Factorization: Alternatively, we use the prime factorization $\mathfrak{p}S = \prod_{i=1}^g \mathfrak{q}_i^{e_i}$. By the Chinese Remainder Theorem, since the powers $\mathfrak{q}_i^{e_i}$ are pairwise comaximal:

$$S/\mathfrak{p}S \cong \bigoplus_{i=1}^{g} S/\mathfrak{q}_i^{e_i}$$

The dimension is additive:

$$\dim_k(S/\mathfrak{p}S) = \sum_{i=1}^{g} \dim_k(S/\mathfrak{q}_i^{e_i})$$

Consider the filtration of $S/\mathfrak{q}_i^{e_i}$ given by the powers of $\mathfrak{q}_i$:

$$S \supset \mathfrak{q}_i \supset \mathfrak{q}_i^2 \supset \cdots \supset \mathfrak{q}_i^{e_i}$$

Each successive quotient $V_j = \mathfrak{q}_i^j/\mathfrak{q}_i^{j+1}$ is a vector space over the field $S/\mathfrak{q}_i$. Since $\mathfrak{q}_i$ is invertible in the Dedekind domain S, there are no proper ideals between $\mathfrak{q}_i^j$ and $\mathfrak{q}_i^{j+1}$, implying V_j is a one-dimensional vector space over $S/\mathfrak{q}_i$.

Thus, as a k-vector space:

$$\dim_k(\mathfrak{q}_i^j/\mathfrak{q}_i^{j+1}) = [S/\mathfrak{q}_i : k] \cdot \dim_{S/\mathfrak{q}_i}(V_j) = f_i \cdot 1 = f_i$$

Summing over the chain of length e_i, we get $\dim_k(S/\mathfrak{q}_i^{e_i}) = e_i f_i$.

Equating the two calculations of the dimension yields $n = \sum_{i=1}^g e_i f_i$. □

Now that we understand what primes might do, it is natural to ask how we might figure out what a specific prime actually does in more detail. There are many tools for this which we will investigate in these sections. The most immediately natural one is the *Dedekind-Kummer* theorem.

Theorem 2.17. *Let R be a Dedekind domain with fraction field K, and $L = K(\alpha)$ a finite separable extension with $\alpha \in S$, the integral closure of R in L. Let $f(x) \in R[x]$ be the minimal polynomial of α.*

If $\mathfrak{p}$ is a prime ideal of R such that there is a natural isomorphism:

$$\phi : (R/\mathfrak{p})[x]/(\bar{f}(x)) \xrightarrow{\sim} S/\mathfrak{p}S \tag{1}$$

Then the factorization of $\bar{f}(x)$ into monic irreducibles in $(R/\mathfrak{p})[x]$ as $\bar{f}(x) = \bar{g}_1(x)^{e_1} \cdots \bar{g}_k(x)^{e_k}$ controls the behaviour of $\mathfrak{p}$ in the following way. Let $g_i(x)$ be a monic lift of $\bar{g}_i(x)$ to $R[x]$. Then $\mathfrak{p}S$ factors into prime ideals in S as $\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_k^{e_k}$, where $\mathfrak{q}_i = (\mathfrak{p}, g_i(\alpha))$. Furthermore, the inertia degree is $f(\mathfrak{q}_i|\mathfrak{p}) = \deg(\bar{g}_i)$.

Proof. We utilize the assumed structural isomorphism to transport the factorization from the polynomial ring to the Dedekind domain S.

Algebraic Decomposition: Since the factors $\bar{g}_i(x)$ are distinct and irreducible in $(R/\mathfrak{p})[x]$, they are pairwise coprime. Applying the Chinese Remainder Theorem to the quotient ring $(R/\mathfrak{p})[x]/(\bar{f}(x))$ yields the decomposition:

$$\frac{(R/\mathfrak{p})[x]}{(\bar{f}(x))} \cong \prod_{i=1}^{k} \frac{(R/\mathfrak{p})[x]}{(\bar{g}_i(x)^{e_i})}$$

Via the isomorphism ϕ, we obtain the corresponding structure for the quotient of the integral closure:

$$S/\mathfrak{p}S \cong \prod_{i=1}^{k} \frac{(R/\mathfrak{p})[x]}{(\bar{g}_i(x)^{e_i})}$$

Ideal Correspondence: The prime ideals of the ring S containing $\mathfrak{p}S$ correspond bijectively to the prime ideals of the quotient ring $S/\mathfrak{p}S$. Under the decomposition above, the maximal ideals of the product ring are generated by the classes of $\bar{g}_i(x)$.

Let $\mathfrak{q}_i$ be the ideal in S generated by $\mathfrak{p}$ and $g_i(\alpha)$. Under ϕ, the image of $\mathfrak{q}_i$ in $S/\mathfrak{p}S$ corresponds to the ideal $(\bar{g}_i(x))$. Since $(R/\mathfrak{p})[x]/(\bar{g}_i(x))$ is a field (as $\bar{g}_i$ is irreducible), the ideal $\mathfrak{q}_i$ is maximal in S. Thus, $\mathfrak{q}_1, \ldots, \mathfrak{q}_k$ are the distinct prime ideals lying over $\mathfrak{p}$.

Factorization and Degrees: The local factor $(R/\mathfrak{p})[x]/(\bar{g}_i(x)^{e_i})$ is a local ring with a unique maximal ideal which is nilpotent of index e_i. This implies that in the factorization of $\mathfrak{p}S$, the prime ideal $\mathfrak{q}_i$ must appear with exponent e_i. Therefore, $\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_k^{e_k}$.

Finally, the inertia degree is given by the dimension of the residue field extension:

$$f(\mathfrak{q}_i|\mathfrak{p}) = [S/\mathfrak{q}_i : R/\mathfrak{p}] = \dim_{R/\mathfrak{p}} \left(\frac{(R/\mathfrak{p})[x]}{(\bar{g}_i(x))} \right) = \deg(\bar{g}_i)$$

□

While this theorem is very strong, and in particular tells us exactly how each prime behaves, the condition in Eq. (1) is very opaque. We will therefore spend a lot of time talking about how to actually see whether this is true in later sections.

2.3. Ideal Norms

In this section, I will define some machinery that will allow us to reason more smoothly with ideals. That is the ideal norm.

Remark 2.18. If R is a PID recall Remark 2.14 that S is an R-lattice. In this case multiplication by some $s \in S$ sends this lattice to a sub-lattice of itself via some R-matrix. The determinant of this matrix is by definition the norm of s (since the lattice is spanned by a K-basis of L).

By looking at the image of the lattice under s, we can detect the norm of s up to a unit (since any matrix with the same image has the same determinant). Essentially, we can see the ideal that the norm of s is generating. This motivates the following generalization of the concept to ideals.

Definition 2.19 (Ideal Norm)**.** Suppose that R is a PID and that I is an ideal of S. Then I is a sub-lattice of S. We define the *ideal norm* of I as the ideal generated by the determinant of any matrix sending S to I as R-lattices.

Proof. There are a few things to check:

- This definition works since R being a PID does indeed make S an R-lattice;
- Since S is full rank in L any two matrices sending S to I have the same determinant up to a unit.
- The ideal norm is never 0 since we may find some $x \in K$ such that $S \subseteq xI$. Therefore the sub-lattice must have full rank.

□

The most important example of this is when $R = \mathbb{Z}$. In this case the determinant of the transition matrix is just the index of the sub-lattice:

$$\mathrm{N}(I) = [S : I]$$

Unfortunately when R is not a PID this definition break down. Luckily the localization of a Dedekind domain is a PID (indeed, it is a DVR). Therefore we can compute the valuations of the ideal norm at each prime, and use that to characterize the full ideal norm.

Remark 2.20. The ideal norm agrees with the element norm on principal ideals:

$$\mathrm{N}((a)) = (\mathrm{N}(a))$$

Proof. The matrix of the action of some $x \in S$ is a matrix sending S to (s). $\square$

Theorem 2.21. *The ideal norm is multiplicative. For ideals $\mathfrak{a}, \mathfrak{b} \subseteq S$:*

$$\mathcal{N}_{L/K}(\mathfrak{a}\mathfrak{b}) = \mathcal{N}_{L/K}(\mathfrak{a})\mathcal{N}_{L/K}(\mathfrak{b})$$

Proof. Working locally, we may assume that R is a DVR, therefore S is semi-local and a PID. Then for ideals (a) and (b) the result follows from the multiplicativity of the element norm

$$\mathrm{N}(ab) = \mathrm{N}(a)\,\mathrm{N}(b)$$

and Remark 2.20. $\square$

Theorem 2.22. *The ideal norm is explicitly*

$$\mathrm{N}(\mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_n^{e_n}) = \prod_i (R \cap \mathfrak{q}_i)^{e_i f_{\mathfrak{q}_i}}$$

where $f_{\mathfrak{q}_i}$ is the inertial degree of $\mathfrak{q}_i$.

Proof. By Theorem 2.21 it suffices to prove that $\mathrm{N}(\mathfrak{q}) = \mathfrak{p}^f$ where f is the inertia of $\mathfrak{q}$ and $\mathfrak{p} = \mathfrak{q} \cap R$. Working locally let $\mathfrak{q} = (\lambda)$, $\mathfrak{p} = (\pi)$, and R be a DVR. Write the exact sequence

$$S \underset{m_\lambda}{\to} \mathfrak{q} \to S/\mathfrak{q} \to 0$$

where m_λ is the matrix describing the multiplication by λ. By changing basis, we may put m_π into *Smith normal form* so that:

$$m_\lambda = \begin{bmatrix} \pi^{c_1} & 0 & \cdots & 0 & \\ 0 & \pi^{c_2} & \cdots & 0 & \\ \vdots & \vdots & \ddots & & \\ 0 & 0 & 0\cdots & 0 & \pi^{c_k} \end{bmatrix} \qquad S/\mathfrak{q} \cong \prod_{i=1}^{k} R/\mathfrak{p}^{c_i}$$

From this expression we see that

$$\det m_\lambda = (\pi)^{c_1+\cdots+c_k} = \mathfrak{p}^{c_1+\cdots+c_k} \qquad f_{\mathfrak{q}} = \dim_{R/\mathfrak{p}} S/\mathfrak{q} = c_1 + \cdots + c_k$$

as desired. $\square$

It is worth noting that all this machinery also works for fractional ideals just as well.

2.4. The Different Ideal

In this section, we will investigate futher prime ramification. We will need the assumption that $R/\mathfrak{q}$ is a perfect field (which is always true for number fields), in order to proceed.

Lemma 2.23 (Nilpotents force degeneracy)**.** *Let k be a field and A a finite-dimensional* commutative *k-algebra. If A has a nontrivial nilpotent element then the trace pairing $\mathrm{Tr}_{A/k}$ is degenerate.*

Proof. If $x \in A$ is nilpotent then xy is nilpotent for every y, hence has trace 0. Thus $\mathrm{Tr}_{A/k}(xy) = 0$ for all y, i.e. x lies in the left kernel of $\mathrm{Tr}_{A/k}$. $\square$

Definition 2.24 (Discriminant ideal)**.** The *discriminant ideal* $\mathrm{disc}(S/R) \subseteq R$ is the invertible ideal characterized by the local rule: for every nonzero prime $\mathfrak{p} \subset R$ and every $R_\mathfrak{p}$-basis $e_1, \ldots, e_n$ of $S_\mathfrak{p}$,

$$\mathrm{disc}(S/R)_\mathfrak{p} \;=\; \big(\det(\mathrm{Tr}_{L/K}(e_i e_j))\big) \subseteq R_\mathfrak{p}.$$

If R is a PID (for instance $\mathbb{Z}$) then S is a free R-module and we can use Definition 2.7 to compute $\operatorname{disc}(S/R)$. However this is not in general true so we might run into problems. On the other hand the localizations of R are DVRs, so each $S_\mathfrak{p}$ is indeed free over $R_\mathfrak{p}$ (by the structure theorem for fingen modules over PIDs) and we can compute the discriminant locally.

Lemma 2.25. *Suppose that $\mathfrak{p}$ is a prime of R with $\mathfrak{q} \in \operatorname{Spm} \mathfrak{q}$ lying over it. Then the trace pairing controls ramification in the following way.*

$$\begin{array}{ccccc} \mathfrak{p} \textit{ ramifies} & \Leftrightarrow & S/\mathfrak{p} \times S/\mathfrak{p} \to R/\mathfrak{p} \textit{ degenerate} & \Leftrightarrow & S_\mathfrak{p} \times S_\mathfrak{p} \to R_\mathfrak{p} \textit{ degenerate} \\ \mathfrak{q} \textit{ ramifies} & \Leftrightarrow & S_\mathfrak{q}/\mathfrak{p} \times S_\mathfrak{q}/\mathfrak{p} \to R/\mathfrak{p} \textit{ degenerate} & \Leftrightarrow & S_\mathfrak{q} \times S_\mathfrak{q} \to R_\mathfrak{p} \textit{ degenerate} \end{array}$$

Proof. Write $\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_g^{e_g}$. Now by Chinese Remainder theorem

$$S/\mathfrak{p}S \cong \bigoplus_{i=1}^{g} S/\mathfrak{q}_i^{e_i}.$$

The summand $S/\mathfrak{q}_i^{e_i}$ is reduced iff $e_i = 1$, and has nilpotents iff $e_i > 1$. Therefore $S/\mathfrak{p}S$ is reduced iff all $e_i = 1$. Then by Lemma 2.23 ramification makes the reduced trace pairing $\operatorname{Tr}_{(S/\mathfrak{p})/(R/\mathfrak{p})}$ degenerate. In the other direction if $e_i = 1$ then $S/\mathfrak{p}S$ is a direct sum of separable extensions of $R/\mathfrak{p}$ (since $R/\mathfrak{p}$ is assumed perfect). Therefore $\operatorname{Tr}_{(S/\mathfrak{p})/(R/\mathfrak{p})}$ is perfect on each summand so it is perfect on $S/\mathfrak{p}S$.

Taking the localization at some $\mathfrak{q}_k$ isolates a single term of the summand. Our previous argument finishes the theorem for $\operatorname{Tr}_{(S_\mathfrak{q}/\mathfrak{p})/(R/\mathfrak{p})}$

To obtain the second column, note that taking the discriminant commutes with quotient and localization. Then since $R_\mathfrak{p}$ is a DVR the discriminant being a non unit in the localization is equivalent to being 0 in the quotient. ☐

This gives us a very interesting connection. Prime ramification comes from the failure of the trace pairing to be perfect. Therefore we try to understand when this occurs.

Construction 2.26 (Trace-duality map)**.** The trace pairing on L/K restricts to an R-bilinear pairing on S, and hence an R-linear map

$$\tau_{S/R} : S \longrightarrow S^\vee := \operatorname{Hom}_R(S, R), \qquad x \longmapsto \langle x, - \rangle.$$

Over K we have shown that the trace duality is an isomorphism $L \xrightarrow{\sim} \operatorname{Hom}_K(L, K)$ because L/K is separable. However over R the pairing may fail to be perfect. In particular $\tau_{S/R}$ is injective and $\operatorname{coker}(\tau_{S/R})$ is a torsion R-module. The *different* is a tool that will allow us to understand "how much" this pairing fails to be perfect over R.

Definition 2.27 (Codifferent / trace dual)**.** The *codifferent* (or *trace dual*) of S/R is the fractional S-ideal

$$S^\sharp := \{x \in L : \operatorname{Tr}_{L/K}(xS) \subseteq R\}.$$

as in Construction 2.5.

Remark 2.28. Recall that $S^\sharp$ is identified with $S^\vee$ through the trace-duality map. Then the statement that the pairing is perfect on S is exactly that $S = S^\sharp$ (so that $S \to S^\vee$ is an isomorphism). Unfortunately the map is not neccesarily surjective when restricted to S, so $S^\sharp$ is generally bigger than S. We measure the failure of perfectness by exactly how much bigger it is. However since it's annoying to work with ideals bigger than S, we may as well just invert it.

Definition 2.29 (Different)**.** The *different* of S/R is the inverse fractional ideal

$$\mathfrak{D}_{S/R} := (S^\sharp)^{-1} = \{y \in L : \ yS^\sharp \subseteq S\}.$$

Since $S \subseteq S^\sharp$ (trace of an integral element is integral), we have $\mathfrak{D}_{S/R} \subseteq S$.

Now the failure of perfectness is measured by how much smaller than S the different is.

Proposition 2.30. *The discriminant is the ideal norm of the different.*

Proof. We work locally so that R is a DVR. The ideal norm of $S^\sharp$ is by definition the determinant of the map of lattices sending S to $S^\sharp$. Therefore $\mathrm{N}(\mathfrak{D}_{S/R}) = \mathrm{N}(S^\sharp)^{-1}$ is the determinant of the map of lattices sending $S^\sharp$ to S. This is $\operatorname{disc} S/R$ by Lemma 2.4. $\square$

Theorem 2.31. *Let $\mathfrak{p} \subset R$ be a nonzero prime. The following are equivalent*

(a) $\mathfrak{p} \mid \operatorname{disc}(S/R)$*;*

(b) $S^\sharp_\mathfrak{p} \neq S_\mathfrak{p}$ *(equivalently, $\mathfrak{D}_{S/R,\mathfrak{p}} \neq S_\mathfrak{p}$);*

(c) $\mathfrak{p}$ *is ramified.*

Moreover if $\mathfrak{q} \in \operatorname{Spm} S$ lies over $\mathfrak{p}$, then the following are equivalent

(c) $\mathfrak{q} \mid \mathfrak{D}_{S/R,\mathfrak{q}}$ *(equivalently $S^\sharp_\mathfrak{q} \neq S_\mathfrak{q}$ or $\mathfrak{D}_{S/R,\mathfrak{q}}$);*

(d) $\mathfrak{q}$ *is ramified.*

Proof. By Remark 2.28 (b) is the statement that the localized trace pairing $S_\mathfrak{p} \times S_\mathfrak{p} \to R_\mathfrak{p}$ is degenerate. Morever (a) is the statement that the reduced trace pairing $S/\mathfrak{p} \times S/\mathfrak{p} \to R/\mathfrak{p}$ is degenerate. The equivalence $(a) \Leftrightarrow (b) \Leftrightarrow (c)$ is Lemma 2.25 .

By Remark 2.28 (d) is the statement that the localized trace pairing $S_\mathfrak{q} \times S_\mathfrak{q} \to R_\mathfrak{p}$ is degenerate. Then $(c) \Leftrightarrow (d)$ follows from Lemma 2.25. $\square$

Theorem 2.32 (Monogenic formula for the different)**.** *Suppose $S = R[\alpha]$ for some $\alpha \in S$, and let $f(x) \in R[x]$ be the minimal polynomial of α over K. Then*

$$S^\sharp = \frac{1}{f'(\alpha)}\, S \qquad \textit{and hence} \qquad \mathfrak{D}_{S/R} = \big(f'(\alpha)\big) \subseteq S.$$

In particular the discriminant of the power basis satisfies

$$\operatorname{disc}(1, \alpha, \ldots, \alpha^{n-1}) = (-1)^{\frac{n(n-1)}{2}}\ \mathrm{N}_{L/K}\big(f'(\alpha)\big),$$

where the left-hand side is the determinant $\det(\mathrm{Tr}_{L/K}(\alpha^{i+j-2}))$*.*

Proof. This is the standard "derivative" description of the trace dual in a monogenic extension: the trace-dual basis to $1, \alpha, \ldots, \alpha^{n-1}$ is obtained by dividing appropriate polynomials in α by $f'(\alpha)$, which gives $S^\sharp = (1/f'(\alpha))S$. The discriminant identity then follows by taking norms and comparing with Definition 2.24. $\square$

By Theorem 2.31 a prime $\mathfrak{p} \subset R$ divides $\operatorname{disc}(S/R)$ if and only if some $\mathfrak{q} \mid \mathfrak{p}$ divides $\mathfrak{D}_{S/R}$. This says that the different (resp. discriminant) is supported exactly on the ramified primes of S (resp. of R). In particular, only finitely many primes ramify in a number field extension.

3. The Additive Lattice

Note that $\mathbb{Z}$ is the prototypical example of a Dedekind domain. Given a finite separable field extension $L/\mathbb{Q}$ (in this case L is called a *number field*), Theorem 2.13 tells us that the integral closure of $\mathbb{Z}$ in L, called the *ring of integers* and denoted $\mathcal{O}_L$, is always a Dedekind domain.

3.1. The Minkowski embedding

We have become very familiar with the concept that the ring of integers of a PID in a finite separable extension is a lattice. And moreover that the discriminant of the extension is the covolume of the lattice under the trace pairing. In the case of a number field, our inuition for lattices becomes much more precise.

From now on we fix a number field $K/\mathbb{Q}$ of degree n and write $\mathcal{O} := \mathcal{O}_K$. Let r_1 be the number of real embeddings and r_2 the number of *pairs* of complex conjugate embeddings, so that

$$n = r_1 + 2r_2.$$

Choose embeddings

$$\sigma_1, \dots, \sigma_{r_1} : K \hookrightarrow \mathbb{R} \qquad \text{and} \qquad \tau_1, \dots, \tau_{r_2} : K \hookrightarrow \mathbb{C}$$

where each τ_j is one representative from a conjugate pair (so the full list of embeddings is $\sigma_1, \dots, \sigma_{r_1}, \tau_1, \bar{\tau}_1, \dots, \tau_{r_2}, \bar{\tau}_{r_2}$). We will identify $\mathbb{C} \cong \mathbb{R}^2$ by $x + iy \mapsto (x, y)$.

Construction 3.1 (The additive embedding)**.** Define the map

$$\sigma : K \longrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \cong \mathbb{R}^n, \qquad x \longmapsto \big(\sigma_1(x), \dots, \sigma_{r_1}(x), \tau_1(x), \dots, \tau_{r_2}(x)\big),$$

where we view each $\tau_j(x) \in \mathbb{C}$ as a point in $\mathbb{R}^2$ via real and imaginary parts.

This is an injective homomorphism of additive groups, and it extends $\mathbb{R}$-linearly to an isomorphism $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$.

Proposition 3.2. *The subgroup $\sigma(\mathcal{O}) \subseteq \mathbb{R}^n$ is a lattice. Moreover if $d_K \in \mathbb{Z}$ generates the discriminant ideal* $\operatorname{disc}(\mathcal{O}/\mathbb{Z}) \subseteq \mathbb{Z}$*, then*

$$\operatorname{covol}(\sigma(\mathcal{O})) \;=\; 2^{-r_2}\sqrt{|d_K|}.$$

Proof.

$\sigma(\mathcal{O})$ is a lattice: By Remark 2.14 (with $R = \mathbb{Z}$) we have $\mathcal{O} \cong \mathbb{Z}^n$ as an abelian group. Choose a $\mathbb{Z}$-basis $\omega_1, \dots, \omega_n$ of $\mathcal{O}$. Since σ extends to an $\mathbb{R}$-linear isomorphism $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^n$, the vectors $\sigma(\omega_1), \dots, \sigma(\omega_n)$ are $\mathbb{R}$-linearly independent. Therefore

$$\sigma(\mathcal{O}) = \mathbb{Z}\sigma(\omega_1) \oplus \cdots \oplus \mathbb{Z}\sigma(\omega_n)$$

is a lattice.

Computing the covolume: Let V be the usual *embedding matrix* (as in Proposition 2.8) for the ordered list of all n embeddings $\sigma_1, \dots, \sigma_{r_1}, \tau_1, \bar{\tau}_1, \dots, \tau_{r_2}, \bar{\tau}_{r_2}$:

$$V_{ij} := \sigma_i(\omega_j) \in \mathbb{C}.$$

By Proposition 2.8 we have

$$d_K \;=\; \operatorname{disc}(\omega_1, \dots, \omega_n) \;=\; (\det V)^2.$$

Now let M be the real $n \times n$ matrix whose rows are obtained from V by replacing each conjugate pair of rows

$$(\tau_j(\omega_1), \dots, \tau_j(\omega_n)), \qquad (\bar{\tau}_j(\omega_1), \dots, \bar{\tau}_j(\omega_n))$$

with the two real rows

$$(\operatorname{Re}\tau_j(\omega_1), \dots, \operatorname{Re}\tau_j(\omega_n)), \qquad (\operatorname{Im}\tau_j(\omega_1), \dots, \operatorname{Im}\tau_j(\omega_n)).$$

Concretely, for each j this is a $\mathbb{R}$-linear change of basis on the 2-dimensional row-span given by

$$\begin{pmatrix} \operatorname{Re} z \\ \operatorname{Im} z \end{pmatrix} \longleftrightarrow \begin{pmatrix} z \\ \bar{z} \end{pmatrix}, \qquad \begin{pmatrix} z \\ \bar{z} \end{pmatrix} = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix} \begin{pmatrix} \operatorname{Re} z \\ \operatorname{Im} z \end{pmatrix}.$$

The determinant of $\begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ is $-2i$, so taking all r_2 conjugate pairs we obtain

$$\det V = (\pm(2i)^{r_2}) \det M \qquad \Rightarrow \qquad |\det M| = 2^{-r_2} |\det V|.$$

Finally the covolume of $\sigma(\mathcal{O})$ is exactly $|\det M|$, so

$$\operatorname{covol}(\sigma(\mathcal{O})) = |\det M| = 2^{-r_2} |\det V| = 2^{-r_2} \sqrt{|(\det V)^2|} = 2^{-r_2} \sqrt{|d_K|}.$$

$\square$

Lemma 3.3. *If $I \subseteq \mathcal{O}$ is a nonzero ideal then $\sigma(I) \subseteq \mathbb{R}^n$ is a sublattice of $\sigma(\mathcal{O})$ and*

$$\operatorname{covol}(\sigma(I)) = \operatorname{covol}(\sigma(\mathcal{O})) \cdot [\mathcal{O} : I] = 2^{-r_2} \sqrt{|d_K|} \cdot \mathrm{N}(I).$$

Proof. Since I is a $\mathbb{Z}$-submodule of $\mathcal{O}$ of finite index, its image $\sigma(I)$ is a finite index subgroup of the lattice $\sigma(\mathcal{O})$, hence a sublattice. The index is preserved because σ is injective:

$$[\sigma(\mathcal{O}) : \sigma(I)] = [\mathcal{O} : I].$$

Finally for lattices, finite index is the ratio of covolumes:

$$\operatorname{covol}(\sigma(I)) = \operatorname{covol}(\sigma(\mathcal{O})) \cdot [\sigma(\mathcal{O}) : \sigma(I)]$$

and $[\mathcal{O} : I] = \mathrm{N}(I)$ by the discussion in the ideal norm section (for $R = \mathbb{Z}$). $\square$

3.2. Minkowski's convex body theorem

Theorem 3.4 (Minkowski)**.** *Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice and let $C \subseteq \mathbb{R}^n$ be convex, centrally symmetric ($C = -C$), and measurable. If*

$$\operatorname{vol}(C) > 2^n \operatorname{covol}(\Lambda),$$

then C contains a nonzero lattice point.

Proof. Let $\pi : \mathbb{R}^n \to \mathbb{R}^n/\Lambda$ be the quotient map. The quotient is a real torus of volume $\operatorname{covol}(\Lambda)$, and π is locally volume preserving. Consider $\frac{1}{2}C = \{x/2 : x \in C\}$. Scaling gives $\operatorname{vol}(\frac{1}{2}C) = \operatorname{vol}(C)/2^n$, so the hypothesis implies

$$\operatorname{vol}\Big(\frac{1}{2}C\Big) > \operatorname{covol}(\Lambda) = \operatorname{vol}(\mathbb{R}^n/\Lambda).$$

Therefore $\pi(\frac{1}{2}C)$ cannot be injective (pigeonhole for volume), so we can find distinct $x, y \in \frac{1}{2}C$ with $\pi(x) = \pi(y)$. This means $x - y \in \Lambda$ and $x - y \neq 0$.

Finally, since C is convex and symmetric, we have

$$x, y \in \tfrac{1}{2}C \quad \Rightarrow \quad x - y \in C.$$

Indeed, $x, y \in \frac{1}{2}C$ implies $2x, 2y \in C$, so by symmetry $-2y \in C$ and by convexity

$$(2x) + (-2y) \in C + C \subseteq 2C \qquad \Rightarrow \qquad x - y \in C.$$

Thus C contains the nonzero lattice point $x - y$. $\square$

3.3. The Minkowski bound

We now feed Theorem 3.4 the specific convex body that remembers the field norm.

Lemma 3.5. *For $T > 0$ define*

$$B_T := \left\{ (x_1, \ldots, x_{r_1}, z_1, \ldots, z_{r_2}) \in \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \ : \ \sum_{i=1}^{r_1} |x_i| + 2\sum_{j=1}^{r_2} |z_j| \le T \right\}.$$

Then B_T is convex, symmetric, and

$$\mathrm{vol}(B_T) = \frac{2^{r_1 - r_2}\pi^{r_2}}{n!}\, T^n.$$

Proof.

(a) Convexity and symmetry are immediate from the defining inequality.

(b) For the volume, by symmetry in the real coordinates we may restrict to $x_i \ge 0$ and multiply by 2^{r_1}. For the complex coordinates we use polar coordinates $z_j = \rho_j e^{i\theta_j}$ so $d(\operatorname{Re} z_j)\, d(\operatorname{Im} z_j) = \rho_j\, d\rho_j\, d\theta_j$ and $\int_0^{2\pi} d\theta_j = 2\pi$. Thus

$$\mathrm{vol}(B_T) = 2^{r_1}(2\pi)^{r_2} \int_{\substack{x_i \ge 0,\ \rho_j \ge 0 \\ \sum x_i + 2\sum \rho_j \le T}} \Big(\prod_{j=1}^{r_2} \rho_j\Big)\, dx\, d\rho.$$

Substitute $s_j := 2\rho_j$ so $\rho_j\, d\rho_j = \frac{1}{4} s_j\, ds_j$ and hence $\prod \rho_j\, d\rho_j = 4^{-r_2} \prod s_j\, ds_j$. We get

$$\mathrm{vol}(B_T) = 2^{r_1}(2\pi)^{r_2} 4^{-r_2} \int_{\substack{x_i \ge 0,\ s_j \ge 0 \\ \sum x_i + \sum s_j \le T}} \Big(\prod_{j=1}^{r_2} s_j\Big)\, dx\, ds.$$

Now we integrate out the x_i first. For fixed $s = (s_1, \ldots, s_{r_2})$ with $\sum s_j \le T$, the region in the x-variables is the standard simplex

$$\{x_i \ge 0 : x_1 + \cdots + x_{r_1} \le T - \sum s_j\},$$

which has volume $\frac{(T - \sum s_j)^{r_1}}{r_1!}$. Therefore the integral becomes

$$\int_{\substack{s_j \ge 0 \\ \sum s_j \le T}} \frac{(T - \sum s_j)^{r_1}}{r_1!} \Big(\prod_{j=1}^{r_2} s_j\Big)\, ds.$$

This is a standard simplex integral (one can do it by induction using $\int_0^A (A-t)^a t^b dt = \frac{a!\, b!}{(a+b+1)!} A^{a+b+1}$), and it evaluates to $\frac{T^n}{n!}$. Putting everything together:

$$\mathrm{vol}(B_T) = 2^{r_1}(2\pi)^{r_2} 4^{-r_2} \cdot \frac{T^n}{n!} = \frac{2^{r_1 - r_2}\pi^{r_2}}{n!}\, T^n.$$

$\square$

Theorem 3.6 (Minkowski lemma)**.** *Let $I \subseteq \mathcal{O}$ be a nonzero ideal. Then there exists $0 \ne \alpha \in I$ such that*

$$|\, \mathrm{N}_{K/\mathbb{Q}}(\alpha)| \le \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n} \sqrt{|d_K|} \cdot \mathrm{N}(I).$$

Proof. We apply Theorem 3.4 to the lattice $\Lambda = \sigma(I) \subseteq \mathbb{R}^n$ and the convex body B_T from Lemma 3.5. By Lemma 3.3 we have

$$\mathrm{covol}(\Lambda) = 2^{-r_2}\sqrt{|d_K|} \cdot \mathrm{N}(I).$$

Using Lemma 3.5, the inequality $\mathrm{vol}(B_T) > 2^n\, \mathrm{covol}(\Lambda)$ is

$$\frac{2^{r_1 - r_2}\pi^{r_2}}{n!} T^n > 2^n \cdot 2^{-r_2}\sqrt{|d_K|} \cdot \mathrm{N}(I).$$

Since $n = r_1 + 2r_2$, this rearranges to

$$T^n > \left(\frac{4}{\pi}\right)^{r_2} n! \sqrt{|d_K|} \cdot \mathrm{N}(I).$$

Choose such a T. Then Minkowski gives a nonzero lattice point $\sigma(\alpha) \in \sigma(I) \cap B_T$, i.e. a nonzero $\alpha \in I$ with

$$\sum_{i=1}^{r_1} |\sigma_i(\alpha)| + 2 \sum_{j=1}^{r_2} |\tau_j(\alpha)| \le T.$$

Now apply AM–GM to the n nonnegative numbers

$$|\sigma_1(\alpha)|, \dots, |\sigma_{r_1}(\alpha)|,\ |\tau_1(\alpha)|, |\tau_1(\alpha)|, \dots, |\tau_{r_2}(\alpha)|, |\tau_{r_2}(\alpha)|.$$

Their product is $|\mathrm{N}_{K/\mathbb{Q}}(\alpha)|$ and their average is $\le T/n$, so

$$|\mathrm{N}_{K/\mathbb{Q}}(\alpha)| \le \left(\frac{T}{n}\right)^n.$$

Combining with the inequality for T^n gives

$$|\mathrm{N}_{K/\mathbb{Q}}(\alpha)| \le \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n} \sqrt{|d_K|} \cdot \mathrm{N}(I),$$

as desired. □

Theorem 3.7 (Minkowski bound)**.** *Every ideal class in* Cl_K *contains an integral ideal* $\mathfrak{a} \subseteq \mathcal{O}$ *with*

$$\mathrm{N}(\mathfrak{a}) \le M_K \qquad \textit{where} \qquad M_K := \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n} \sqrt{|d_K|}.$$

Proof. Let $c \in \mathrm{Cl}_K$ and choose an integral ideal $I \subseteq \mathcal{O}$ representing c^{-1}. Apply Theorem 3.6 to I to obtain $0 \ne \alpha \in I$ with

$$|\mathrm{N}_{K/\mathbb{Q}}(\alpha)| \le M_K \cdot \mathrm{N}(I).$$

Since $\alpha \in I$ we have $(\alpha) \subseteq I$, hence

$$\mathfrak{a} := (\alpha) I^{-1} \subseteq \mathcal{O}$$

is an integral ideal. It satisfies $[\mathfrak{a}] = [I^{-1}] = c$, because (α) is principal. Finally by multiplicativity of ideal norm and Remark 2.20 we get

$$\mathrm{N}(\mathfrak{a}) = \mathrm{N}((\alpha))\, \mathrm{N}(I^{-1}) = \frac{|\mathrm{N}_{K/\mathbb{Q}}(\alpha)|}{\mathrm{N}(I)} \le M_K.$$

□

Remark 3.8. It turns out that if we assume the *Generalized Riemann Hypothesis* then the bound in Theorem 3.7 improves substantially. In this case, we can always find a representative $\mathfrak{a}$ of an ideal class with

$$N(\mathfrak{a}) \le 12(\ln |d_K|)^2$$

which is much better than $M_K := \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n} \sqrt{|d_K|}$. In the context of computational algebraic number theory, this is the difference in a polynomial time search and an exponential time search.

3.4. Finiteness of the class number

Now the class number finiteness is essentially a counting statement about sublattices of $\mathbb{Z}^n$.

Lemma 3.9. *For every real number $B > 0$ there are only finitely many integral ideals $\mathfrak{a} \subseteq \mathcal{O}$ with $\mathrm{N}(\mathfrak{a}) \leq B$.*

Proof. Fix a $\mathbb{Z}$-basis $\mathcal{O} \cong \mathbb{Z}^n$. If $\mathfrak{a} \subseteq \mathcal{O}$ is an ideal then it is in particular a $\mathbb{Z}$-submodule, and

$$\mathrm{N}(\mathfrak{a}) = [\mathcal{O} : \mathfrak{a}].$$

For a fixed integer $m \geq 1$, any subgroup $H \subseteq \mathbb{Z}^n$ of index m contains $m\mathbb{Z}^n$ (since m kills the finite group $\mathbb{Z}^n/H$). Therefore such H correspond to subgroups of the finite abelian group

$$\mathbb{Z}^n/m\mathbb{Z}^n \cong (\mathbb{Z}/m\mathbb{Z})^n.$$

A finite group has only finitely many subgroups, so there are only finitely many $\mathbb{Z}$-submodules of index m in $\mathbb{Z}^n$. Finally there are only finitely many integers $m \leq B$, so there are only finitely many ideals with $\mathrm{N}(\mathfrak{a}) \leq B$. ☐

Theorem 3.10. *The class group Cl_K is finite.*

Proof. By Theorem 3.7, every class contains an ideal $\mathfrak{a}$ with $\mathrm{N}(\mathfrak{a}) \leq M_K$. By Lemma 3.9 there are only finitely many such ideals. Therefore there are only finitely many ideal classes. ☐

4. The Units

Construction 4.1. There is a map

$$\iota : \mathcal{O}^\times \to \mathbb{R}^{r_1+r_2} \qquad a \mapsto (\log|\sigma_i(a)|, 2\log|\sigma_j(a)|)$$

where σ_i runs over the real embeddings and σ_j varies over pairs of complex conjugate embeddings. The image is called the *multiplicative lattice*.

Theorem 4.2 (theorem)**.** *The kernel of the map in Construction 4.1 are the roots of unity and the image is a lattice of rank $r_1 + r_2 - 1$.*

Proof. We will extract the properties we want from the properties of the additive lattice.

$\ker \iota = \mathcal{O}^\times$**:** The kernel are the elements such that the absolute value of every embedding is 1. Therefore if σ is the additive embedding then

$$\sigma(a^n) \in [-1, 1]^n$$

however the interestction of a lattice with a box has only finitely many points. Therefore the sequence $a, a^2, \ldots$ contains finitely many elements, therefore $a^n = a^r \implies a^{r-n} = 1$ as desired.

$\operatorname{im} \iota$ **is lattice:** Now we must show the image is a lattice. Since the image is an abelian group it suffices to find a box where only finitely many points live inside it. Note that

$$\{u \in \mathcal{O}^\times : |\iota(u)| \leq \epsilon\} \subseteq \{u \in \mathcal{O}_k : |\sigma(u) \leq e^\epsilon\}$$

which has finitely many points by again considering the additive embedding as desired. To show that it has full rank we want to find an element such that the image has negative values in every coordinate except one. This is due to the following claim

Claim 4.2.1. *If A is a matrix such that $A_{ii} > 0$ and $A_{ij} < 0$ where $i \neq j$ then A has rank $\geq n - 1$.*

Proof. Suppose that we have a linear relation among the first $n - 1$ columns c_i

$$\lambda_1 c_1 + \cdots + \lambda_{n-1} c_{n-1}$$

and assume WLOG that $1 = \lambda_1 > \lambda_i$. Then

$$0 = \sum_k t_i a_{kJ} = a_{kk}$$

☐

Now to find the element b we use the Minkowsi lemma. Essentially the relations $|\sigma_i(b)| < |\sigma_i(a)|$ for $i \neq k$ defines a box that is infinite in one dimension. This box clearly has volume larger than the Minkowsi bound. By constructing a sequence of elements in this box with strictly lower

□

5. Computing in Number Fields

We've done a lot of theoretical thinking and structural work, however we've not yet figured out how to actually compute stuff. In order to do well on an exam, you'll need to learn how to compute stuff. Therefore in this section we'll learn methods for computing things about number fields.

5.1. Finding the Ring of Integers

5.2. Computing Prime Behaviour

5.3. Computing the Class Group

5.4. Finding the Units

6. Cyclotomic Fields

7. The Galois Case

7.1. Decomposition and Inertia Groups

Assume from now on that L/K is a finite *Galois* extension and write

$$G := \mathrm{Gal}(L/K).$$

Let $\mathfrak{p} \subset R$ be a nonzero prime and choose a prime $\mathfrak{q} \subset S$ with $\mathfrak{q} \cap R = \mathfrak{p}$. Write $\kappa(\mathfrak{p}) := R/\mathfrak{p}$ and $\kappa(\mathfrak{q}) := S/\mathfrak{q}$ for the residue fields.

Lemma 7.1. *G acts transitively on the primes above $\mathfrak{p}$.*

Proof. Suppose FTSOC that there are two distinct G orbits of primes above $\mathfrak{p}$. Call one of them them $\{\mathfrak{q}_1, \dots, \mathfrak{q}_n\}$ and let $\{\mathfrak{r}_i\}$ be the primes above $\mathfrak{p}$ not in this orbit. Using CRT we may find $x \in S$ with:

$$x \equiv 1 \pmod{\mathfrak{q}_i} \qquad x \equiv 0 \pmod{\mathfrak{r}_i}$$

Note that $y = \prod_{\sigma \in G} \sigma(x)$ also has these residues and is an element of R. Therefore

$$y \in (R \cap \mathfrak{q}_1) = \mathfrak{p}$$

which is a contradiction with the fact that $y \notin \mathfrak{q}_1$, since it is an ideal that contains $\mathfrak{p}$. □

Corollary 7.2. *Every prime above $\mathfrak{p}$ has the same ramification index and inertia degree. Therefore Theorem 2.16 simplifies to*

$$n = efg$$

where g is the number of primes above $\mathfrak{p}$.

Definition 7.3 (Decomposition group)**.**

- The *decomposition group* of $\mathfrak{q}$ (over $\mathfrak{p}$) is the stabilizer

$$D_{\mathfrak{q}} := \{\sigma \in G : \sigma(\mathfrak{q}) = \mathfrak{q}\} \subseteq G.$$

- Every $\sigma \in D_{\mathfrak{q}}$ induces a $\kappa(\mathfrak{p})$-automorphism of $\kappa(\mathfrak{q})$, hence a homomorphism:

$$\rho_{\mathfrak{q}} : D_{\mathfrak{q}} \longrightarrow \mathrm{Aut}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$$

  The *inertia group* is the kernel:

$$I_{\mathfrak{q}} := \ker(\rho_{\mathfrak{q}}) = \{\sigma \in D_{\mathfrak{q}} : \sigma(x) \equiv x \pmod{\mathfrak{q}} \text{ for all } x \in S\}$$

Proposition 7.4. *The reduction map $\rho_{\mathfrak{q}}$ fits into a short exact sequence*

$$1 \longrightarrow I_{\mathfrak{q}} \longrightarrow D_{\mathfrak{q}} \xrightarrow{\rho_{\mathfrak{q}}} \mathrm{Aut}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p})) \longrightarrow 1$$

Proof. The main task is to show that $\rho_{\mathfrak{q}}$ is surjective. Because $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is finite separable we may choose a primitive element $\bar{\alpha}$ with $\kappa(\mathfrak{q}) = \kappa(\mathfrak{p})[\bar{\alpha}]$. By chinese remainder theorem we may find an element α with

$$\alpha \equiv \bar{\alpha} \pmod{\mathfrak{q}} \qquad \alpha \equiv 0 \pmod{\mathfrak{r}_i} \tag{1}$$

where $\mathfrak{r}_i$ are all the other primes lying over $\mathfrak{p}$. Let

$$f(x) = \prod_{\sigma \in G}(x - \sigma(\alpha)) \qquad \bar{f}(x) = \prod_{\sigma \in G}(x - \overline{\sigma(\alpha)})$$

be the characteristic polynomial of α and its reduction in the residue extension respectively. Since f has coefficients in R, $\bar{f}$ has coefficients in $R/\mathfrak{p}$. Therefore any $\tau \in \mathrm{Aut}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$ sends $\bar{\alpha}$ to another root of $\bar{f}(x)$ and:

$$\tau(\bar{\alpha}) = \overline{\sigma(\alpha)} \qquad \text{for some } \sigma \in G$$

We claim that $\rho_{\mathfrak{q}}(\sigma) = \tau$.

- Suppose for the sake of contradiction that $\sigma \notin D_{\mathfrak{q}}$. Then by Eq. (1) $\sigma(\alpha) \equiv 0 \pmod{\mathfrak{q}}$. Therefore $\overline{\sigma(\alpha)}$ is 0 in the residue extension, a contradiction since τ can't send a nonzero element to 0.
- Since α is primitive, $\tau(\bar{\alpha}) = \overline{\sigma(\alpha)}$ forces $\rho_{\mathfrak{q}}(\sigma_k) = \tau$.

□

Remark 7.5 (The efg-formula realized by subextensions)**.**

(a) $|D_{\mathfrak{q}}| = ef$, $|I_{\mathfrak{q}}| = e$, and $|D_{\mathfrak{q}}/I_{\mathfrak{q}}| = f$.

(b) Considering fixed fields

$$K \subseteq L^D \subseteq L^I \subseteq L$$

  and the degrees in this tower recover the splitting invariants:

$$[L^D : K] = g, \qquad [L^I : L^D] = f, \qquad [L : L^I] = e.$$

Proof.

(a) By orbit stabilizer theorem $|D_{\mathfrak{q}}| = |G|/g$. Then plugging in $|G| = efg$ gives $|D_{\mathfrak{q}}| = ef$ as desired. We must have:

$$ef = |D_{\mathfrak{q}}| = |I_{\mathfrak{q}}||D_{\mathfrak{q}}/I_{\mathfrak{q}}|$$

  Note by Proposition 7.4 that

$$|D_{\mathfrak{q}}/I_{\mathfrak{q}}| = [\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})] = f$$

  which forces $|I_{\mathfrak{q}}| = e$.

(b) This is now a direct translation via the Galois correspondence:

$$[L^D : K] = [G : D_{\mathfrak{q}}] = g, \qquad [L^I : L^D] = [D_{\mathfrak{q}} : I_{\mathfrak{q}}] = f, \qquad [L : L^I] = |I_{\mathfrak{q}}| = e.$$

□

Theorem 7.6 (How to read the tower)**.** *The tower $K \subseteq L^D \subseteq L^I \subseteq L$ packages the three numbers (e, f, g):*

- *L^D/K accounts for the* splitting *(g primes);*
- *L^I/L^D accounts for the* residue field extension *(degree f and unramified);*
- *L/L^I accounts for the* ramification *(index e and totally ramified).*

Proof. Firstly, all the intermediate extensions are Galois since we defined the inertial group without making any choices. Define:

$$\mathfrak{q}_I = \mathcal{O}_{L^I} \cap \mathfrak{q} \qquad \mathfrak{q}_D = \mathcal{O}_{L^D} \cap \mathfrak{q}$$

- Consider the extension L/L^D. Since the Galois group of this extension fixes $\mathfrak{q}$ there are no other primes above $\mathfrak{q}_D$ by Lemma 7.1. Therefore again by Corollary 7.2 there is no splitting in this extension. Now L^D/K must account for all the splitting, and since it has degree g it is totally split.
- Consider the extension L/L^I. By definition the Galois group acts trivially on the residue field extension. On the other hand the Galois group surjects onto the residue field Galois group by the argument in Proposition 7.4. Therefore this extension must have no inertia (and no splitting by previous part). Therefore it is totally ramified, and since it has degree e, it accounts for all the ramification.
- By the previous results L^I/L^D must account for the inertia. Moreover by counting degrees it is totally inert.

□

Theorem 7.7. *Suppose that $K \subseteq E \subseteq L$ is an intermediate extension with Galois group H. Then*

$$I_{\mathfrak{q}}(L/E) = I_{\mathfrak{q}}(L/K) \cap H \qquad D_{\mathfrak{q}}(L/E) = D_{\mathfrak{q}}(L/K) \cap H$$

and moreover if $\mathfrak{q}_E = \mathfrak{q} \cap E$ then there is the following commutative diagram of exact sequences:

$$\begin{tikzcd}
 & 1 \arrow[d] & 1 \arrow[d] & 1 \arrow[d] & \\
1 \arrow[r] & I_{\mathfrak{q}}(L/E) \arrow[r] \arrow[d] & I_{\mathfrak{q}}(L/K) \arrow[r] \arrow[d] & I_{\mathfrak{q}_E}(E/K) \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & D_{\mathfrak{q}}(L/E) \arrow[r] \arrow[d] & D_{\mathfrak{q}}(L/K) \arrow[r] \arrow[d] & D_{\mathfrak{q}_E}(E/K) \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & \overline{G}_{\mathfrak{q}}(L/E) \arrow[r] \arrow[d] & \overline{G}_{\mathfrak{q}}(L/K) \arrow[r] \arrow[d] & \overline{G}_{\mathfrak{q}_E}(E/K) \arrow[r] \arrow[d] & 1 \\
 & 1 & 1 & 1 &
\end{tikzcd}$$

where we have used $\overline{G}$ to denote the automorphisms of the respective residue field.

Proof. Recall the inertial and decomposition groups are defined as subsets of the Galois group without reference to the base field. Therefore restricting to a subset of the Galois group just intersects.

Now consider the commutative diagram. Each column follows from Proposition 7.4. Each row corresponds to an inclusion and then a restriction, in which the inclusion is precisely the kernel of the restriction by definition. The surjectivity roughly comes from the fact that we can embed a field into a larger field in stages. This gives the following bijection

$$\phi : \mathrm{Hom}_K(L, \Omega) \longleftrightarrow \mathrm{Hom}_E(L, \Omega) \times \mathrm{Hom}_K(E, \Omega)$$

which we can view as a bijection of Galois groups, so that the restriction becomes projection onto the second term on the RHS. If $\sigma \in \mathrm{Gal}(E/K)$ stabilizes $\mathfrak{q}_E$ then $\phi^{-1}(1, \sigma) \in \mathrm{Gal}(L, K)$ stabilizes $\mathfrak{q}$ and restricts to σ. This shows surjectivity on the first two rows, and the third follows either from a similar argument, or the nine lemma. □

Remark 7.8. The most natural way to remember this is that:

- L^D is a largest subextension of L/K where $\mathfrak{p}$ is totally split.
- L^I is the largest subextension of L/K where $\mathfrak{p}$ is unramified.

Proof. Suppose that $K \subseteq K' \subseteq L$ is an intermediate extension and consider L/K'. Note Theorem 7.7 tells us that the inertial and decomposition groups are intersections with $\mathrm{Gal}(L/K')$. Since group intersection corresponds to field multiplication in the Galois correspondance we obtain the diagram:

$$\begin{tikzcd}
L \arrow[dd, dash, "e"'] \arrow[dr, dash, "e'"] & \\
& L^I K' \arrow[dd, dash, "f'"] \\
L^I \arrow[dd, dash, "f"'] \arrow[ur, dash] & \\
& L^D K' \arrow[dd, dash, "r'"] \\
L^D \arrow[dd, dash, "r"'] \arrow[ur, dash] & \\
& K' \\
K \arrow[ur, dash] &
\end{tikzcd}$$

Since e, f, g are multiplicative:

- K'/K being totally split is is exactly that $e' = e$ and $f' = f$. This occurs when $L^D K' = L^D$ or equivalently $K' \in L^D$.
- K'/K being unramified is is exactly that $e' = e$. This occurs when $L^I K' = L^I$ or equivalently $K' \in L^I$.

□

While this stuff is extremely beautiful, it is also very useful even when things aren't Galois. For instance consider how this understanding enables the following argument.

Theorem 7.9. *Let L_1 and L_2 be extensions of K. Then:*

- *If $\mathfrak{p}$ is unramified in L_1/K and L_2/K it is unramified in the compositum L_1L_2/K.*
- *If $\mathfrak{p}$ is totally split in L_1/K and L_2/K it is totally split in the compositum L_1L_2/K.*

Proof. Let L be the normal closure of L_1L_2 so that L/K is Galois. By Remark 7.8:

- If $\mathfrak{p}$ is unramified in both extensions then $L_1, L_2 \subseteq L^I$. Therefore $L_1L_2 \subseteq L^I$ exactly meaning $\mathfrak{p}$ is unramified in the compositum.
- If $\mathfrak{p}$ is totally split in both extensions then $L_1, L_2 \subseteq L^D$. Therefore $L_1L_2 \subseteq L^D$ exactly meaning $\mathfrak{p}$ is totally split in the compositum.

□

Remark 7.10. Suppose that $K = \mathbb{Q}$. Then the residue field $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is of the form $\mathbb{F}_{p^n}/\mathbb{F}_p$ for some prime p. Now

$$D_{\mathfrak{q}}/I_q \cong \mathrm{Gal}(L^I/L^D) \cong \mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$$

is a cyclic group of order n generated by the *Frobenius element* $x \mapsto x^p$ on the right hand side. In particular when $\mathfrak{p}$ is unramified then $I_q = 1$ and we may find a unique element $\sigma \in D_{\mathfrak{q}}$ with

$$\sigma(\alpha) \equiv \alpha^p \pmod{\mathfrak{q}}$$

The Frobenius element is extremely useful, and will make many appearences when studying more number theory. As an extra treat, here is a really nice proof of quadratic reciprocity using this theory.

Theorem 7.11 (Reciprocity)**.** *Let p and q be distinct odd primes. Then:*

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}$$

Proof. Let $K = \mathbb{Q}[\zeta_p]$ where ζ_p is a primitive pth root of unity. Since

$$\mathrm{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times \cong \mathbb{Z}/(p-1)\mathbb{Z}$$

it contains a unique subgroup of order $(p-1)/2$ containing the elements of $(\mathbb{Z}/p\mathbb{Z})^\times$ that are squares. Therefore K contains a unique quadratic extension F. On the other hand, only p is ramified in $K/\mathbb{Q}$, so F must have this same property. Now (INSERT EXERCISE) allows us to deduce F:

- If $p \equiv 1 \pmod 4$ then p is the only prime ramifying in $\mathbb{Q}[\sqrt{p}]$ and this is the only quadratic field for which this is true.
- If $p \equiv 3 \pmod 4$ then $-p$ is the only prime ramifying in $\mathbb{Q}[\sqrt{-p}]$ and this is the only quadratic field for which this is true.

Therefore $F = \mathbb{Q}[\sqrt{d}]$ where $d = (-1)^{(p-1)/2} \cdot p$. Since the Galois group of F has order 2 we identify it with $\{+1, -1\}$. Set q to be an odd prime not equal to p and $\tau \in \mathrm{Gal}(K/\mathbb{Q}) = \{+1, -1\}$ to be the Frobenius element with:

$$\tau\alpha \equiv \alpha^q \pmod q \implies \tau : \zeta \mapsto \zeta^q$$

Therefore τ restricts to the identity element of $\mathrm{Gal}(F/\mathbb{Q})$ according to whether q is a square in $(\mathbb{Z}/p\mathbb{Z})^\times$ or not. This exactly means $\tau = \left(\frac{q}{p}\right)$. On the other hand the Frobenius element generates the inertial group, which is nontrivial in our case if and only if q remains inert in F. This corresponds to $\tau = \left(\frac{d}{q}\right)$ again by (INSERT EXERCISE). Now by equating

$$\left(\frac{q}{p}\right) = \left(\frac{-1}{q}\right)^{(p-1)/2} \cdot \left(\frac{p}{q}\right) = (-1)^{\frac{(p-1)(q-1)}{4}} \cdot \left(\frac{p}{q}\right)$$

where we have used that -1 is a square iff $q \equiv 1 \pmod 4$. □

8. Exercises

References

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